聚乙烯物料安全数据表

Material Safety Data Sheet Of Polyethylene

“MARLEX®”、“三人®”商标的产品及相关产品

Product and Relevant Product Under Marlex or Three-man Brand

第一部分　化学品及企业标识

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

俗名：　塑料

Synonyms:　*Plastic*

化学名称：　乙烯均聚物或乙烯/己烯-1 共聚物及相应添加剂

Chemical Name: Ethene, Homopolymer;1-Hexene,Polymer with Ethene; thylene Polymers & Copolymers plus Additives.

化学族：　烯烃均聚物或共聚物及相应添加剂

Chemical Family: Olefin Polymers and Copolymers plus Additives

化学式：　混合物

Chemical Formula: Mixture

CAS 号：　（参见---第二部分　成分/组成信息）

CAS Reg. No.: Mixture-See Section B for ingredient CAS Reg. No.

产品牌号：　HHMTR140、HHMTR144、HXMTR168、HHMTR210、HHMTR230、H401、HHMTR400、HHMTR480、HXM50100CA、HXMTR550、HXMTR570、HHM5202、HHM5502、HHM5503、EHM6007 等，及其它相关牌号的系列产品。

Product No.:

HHMTR140、HHMTR144、HXMTR168、HHMTR210、HHMTR230、H401、HHMTR400、HHMTR480、HXM50100CA、HXMTR550、HXMTR570、HHM5202、HHM5502、HHM5503、EHM6007 etc,and the product of any other types which being correlated.

公司：　上海金菲石油化工有限公司

Company:　*Shanghai Golden Phillips petrochemical Company Limited*

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应急电话(Emergency Tel):

上海化学事故应急咨询服务电话：021-62679090

Shanghai Chemical Accident Emergency Consulting Service Tel.:

上海化学品安全监管电话：　021-62679090

Shanghai Chemical Safety Supervision Tel.:

国家化学事故应急咨询电话：　0532-3889090

National Chemical Accident Emergency Consulting Tel:

产品用途：

PRODUCT USE:

医疗用途警告：不得将上海金菲石油化工有限公司生产的此种产品永久性植入人体或用于其它长期与人体内部体液或组织接触的医疗用途。除非与上海金菲石油化工有限公司签署合同对预期用途加以明确规定并由上海金菲石油化工有限公司直接供应产品，不得将上海金菲石油化工有限公

司生产的此种产品短期或者短暂地植入人体或用于其他短期或者短暂与人体内部体液和组织接触的医疗用途。

MEDICAL APPLICATION CAUTION:

Do not use this ShangHai Golden Phillips Chemical material in medical applications involving permanent implantation in the human body or permanent contact with internal body fluids or issues..Do not use this ShangHai Golden Phillips Chemical material in medical applications involving brief or temporary implantation in the human body or contact with internal body fluids or tissues unless the material has been provided directly from ShangHai Golden Phillips Chemical under a contract which expressly acknowledges the contemplated use.

针对本产品用于植入人体或与人体内部体液或组织接触的适用性，上海金菲石油化工有限公司不作任何陈述、承诺，不作任何明示或暗示的保证。

ShangHai Golden Phillips Chemical makes no representation, promise, express warranty or implied warranty concerning the suitability of this material for use in implantation in the human body or in contact with internal body fluid or tissues.

产品或产品成分是否已列入美国环保总署（EPA）的 TSCA 目录：是

Product and/or Components Entered on EPA's TSCA Inventory: Yes

本产品在美国国内贸易时列入化学品的有毒物质控制法规（TSCA）目录，因为它有可能受到适用的 TSCA 规定和限制的管辖。

This product is in U.S. commerce, and is listed in the Toxic Substances Control Act (TSCA) Inventory of Chemicals; hence, it may be subject to applicable TSCA provisions and restrictions.

列入加拿大目录的情况：DSL

Canadian Inventory Listing Status: DSL

所有的成分均被列入加拿大国内物质目录(DSL)。

All ingredients are listed in the Domestic Substances List (DSL).

根据加拿大环境保护法规（CEPA）第 3 节的规定，所有的杂质均受到豁免。

Impurities are exempt in accordance with Section 3 of the Canadian Environmental Protection Act (CEPA).

第二部分　成分/组成信息

2. COMPOSITION/INFORMATION ON INGREDIENTS

成　份	CAS 号	WT%	OSHA PEL*	ACGIH TLV*
乙烯均聚物	9002-88-4	>96	NE	NE
或				
乙烯-己烯-1 共聚物	25213-02-9	>96	NE	NE
及				
添加剂	多种	<4	NE	NE

备注：（OSHA：美国职业安全与卫生条例；ACGIH：美国政府和工业卫生委员会。）

CAS % OSHA ACGIH

Ingredients	*Number By*	*Wt.*	*PEL**	*TLV**
Polyethylene	*9002-88-4*	*>96*	*NE*	*NE*
or				
Ethylene-Hexene-1 Copolymer	*25213-02-9*	*>96*	*NE*	*NE*
and				
Additives Various	*多种*	*<4*	*NE*	*NE*

第三部分　危险性概述

3. HAZARDS IDENTIFICATION

紧急情况概述：透明、半透明的无色丸状或薄片状物质，在高温下会生成甲醛。

EMERGENCY OVERVIEW:Opaque, translucent, colorless pellets or fluff ***- Formaldehyde can be produced at elevated temperature.***

建议的接触限值：按其它未经规定或分类的颗粒物（PNOC） 进行控制。

OSHA	PEL
可吸入颗粒浓度	5 mg/m^3
粉尘总浓度	15 mg/m^3

Recommended Exposure Limits:Control as Particulate Not Otherwise Classified (PNOC) or Regulated:

OSHA	PEL
Respirable Fraction	5 mg/m^3
Total Dust	15 mg/m^3

备注：　（该数值是不含石棉以及结晶二氧化硅含量 <1% 的（总的）可吸入颗粒物的量。）根据目前可以获得的资料，本产品不符合任何由法规 29 CFR 第 1910.1200 节定义的危险性。

** The value is for inhalable (total) particulate matter containing no asbestos and <1% crystalline silica.._X_ Based on information presently available, this product does not meet any of the hazard definitions of 29 CFR Section 1910.1200.*

明确确定:　加拿大“健康危险品分类”中没有任何已知的可以适用的资料。

Specify: No known applicable information for Canadian "Health Hazard Categories".

其它:　在热加工过程中形成的热分解产物会刺激眼睛。
在挤出温度下有可能生成甲醛，这是一种可使人类致癌的物质。

Other: Thermal decomposition products formed during thermal processing are expected to be irritating to the eyes.

Formaldehyde, A probable human carcinogen, may be formed at extrusion temperatures.

直接的健康影响：

IMMEDIATE HEALTH EFFECTS:

眼：　预期不会造成长期或严重的眼睛刺激。粉尘会引起机械性刺激。如果眼睛接触加热的物料后，会导致热灼伤。

EYE: Not expected to cause prolonged or significant eye irritation. Dust may cause mechanical irritation. If this material is heated, thermal burns may result from eye contact.

皮肤：　与皮肤接触后不会造成长期或明显的刺激。如果经皮肤吸收，不会损害体内器官。如果皮肤接触加热后的物料，会导致热灼伤。

SKIN:Contact with the skin is not expected to cause prolonged or significant irritation. Not expected to be harmful to internal organs if absorbed through the skin. If this material is heated, thermal burns may result from skin contact.

吸入：　如果吸入，不会有害。如果本物料加热后，烟雾会令人不适，并引起恶心，会刺激上呼吸道。粉尘会对鼻粘膜、喉咙以及上呼吸道产生机械性刺激。长期暴露在高粉尘浓度的环境下会造成非衰老性肺病变。

INHALATION:Not expected to be harmful if inhaled. If this material is heated, fumes may be unpleasant and produce nausea and irritation of the upper respiratory tract. Inhalation: Dust may produce mechanical irritation to the mucous membranes of the nose, throat and upper respiratory tract.Long term exposure to high dust concentrations may cause non-debilitating lung changes.

摄入: 如果吞服，不会引起伤害。

INGESTION:Not expected to be harmful if swallowed.

接触引起的体症及症状:

SIGNS AND SYMPTOMS OF EXPOSURE:

对眼睛的热灼伤: 可包括疼痛、流泪、红眼睛、肿胀、视觉减退。

Thermal burns to the eye: may include pain, tearing, reddening, swelling, and impaired vision. Respiratory

对皮肤的热灼伤：可包括疼痛或灼热感、皮肤变色、肿胀以及起水泡。

Thermal burns to the skin:May include pain or feeling of heat, discoloration, swelling, and blistering.

呼吸道刺激：可包括咳嗽和呼吸困难。

Irritation: May include coughing and difficulty breathing.

在挤出温度下（>350F, >177℃），聚乙烯会释放出蒸气和气体，会刺激眼睛的粘膜、口腔、咽喉和肺脏。这些物质可能包括乙醛、丙酮、乙酸、甲酸、甲醛和丙烯醛。通常这些刺激性效应都只是短暂的。但是，长期暴露在刺激性的废气下会导致肺水肿。充分通风应能防止感官不适。根据动物实验数据和有限的流行病学证据，美国国家毒理学研究计划（NTP）、国际癌症研究中心（IARC(2A)）以及职业安全和卫生管理局（OSHA）已经将甲醛列为可能的人类致癌物质。

At extrusion temperatures (>350F, >177℃), polyethylenes can release vapors and gases which are irritating to the muccus membranes of the eyes, mouth,throat, and lungs. These substances may include acetaldehyde, acetone,acetic acid, formic acid, formaldehyde and acrolein. Generally these irritant effects are all transitory. However, prolonged exposure to irritating off gases can lead to pulmonary edema. Adequate ventilation should prevent sensory discomfort. Based on animal data and limited epidemiological evidence,NTP, IARC(2A), and OSHA have listed Formaldehyde as a probable human carcinogen.

第四部分 急救措施

4. FIRST AID MEASURES

眼睛： 如果加热的物料溅入眼睛中，则应立即将眼睑张开，用清水冲洗眼睛 15 分钟。如果佩戴隐形眼镜，则应取下隐形眼镜。立即送医就诊。

EYE:If heated material should splash into eyes, flush eyes immediately with fresh water for 15 minutes while holding the eyelids open. Remove contact lenses, if worn. Get immediate medical attention.

皮肤： 如果热的物料粘附到皮肤上，应迅速用水冷却。如有严重烫伤，应送医就诊。不得将凝固的物料从皮肤上剥下来或用溶剂或稀释剂将其溶解。从被粘附的皮肤上去除该物料时，建议使用植物油或矿物油。

SKIN:If the hot material gets on skin, quickly cool in water. See a doctor for extensive burns. Do not try to peel the solidified material from the skin or use solvents or thinners to dissolve it. The use of vegetable oil or mineral oil is recommended for removal of this material from the skin.

吸入： 吸入者转移到空气新鲜处。如吸入者停止呼吸，施以人工呼吸术。如果呼吸困难，则须供给氧气。如果出现持续性呼吸困难，送医就诊。

INHALATION:Move the exposed person to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical attention if breathing difficulties continue.

摄入： 由于摄入该物质不会引起伤害，所以无需采取任何急救措施。

INGESTION:No specific first aid measures are required because this material is not expected to be harmful if swallowed.

第五部分 消防措施

5. FIRE FIGHTING MEASURES

火灾分类：

FIRE CLASSIFICATION:

分类 （29 CFR 1910.1200）：不属于易燃性或可燃性物质。

Classification (29 CFR 1910.1200): Not flammable or combustible.

易燃性质：

FLAMMABLE PROPERTIES:

闪点： 645F (340℃) ASTM (D1929)。

FLASH POINT: 645F (340C) ASTM (D1929)

自燃温度： 716F (380℃)。

AUTOIGNITION: 716F (380C)

可燃性极限 （% 体积，在空气中）： 下限： 不适用；上限： 不适用。

FLAMMABILITY LIMITS (% by volume in air): Lower: NA Upper: NA

灭火介质： C02、干粉、泡沫及水雾。

EXTINGUISHING MEDIA:C02, dry chemical, foam and water fog

美国防火协会（NFPA）评级： 健康 1； 易燃性 1； 反应性 0。

NFPA RATINGS: Health 1; Flammability 1; Reactivity 0.

灭火说明： 虽然本物质不易点燃，但可燃烧。如果火灾现场储有本物质，在未配带包括自给式呼吸器（NIOSH SCBA）等适用的防护设备之前不得进入火灾现场内任何封闭或受限空间。由于这是一种表面燃烧的物质，如有可能应当用一个雾化喷枪喷施水雾。施以高速度的灭火水柱，有可能会扩大燃烧表面层。

FIRE FIGHTING INSTRUCTIONS:This material will burn although it is not easily ignited. For fires involving this material, do not enter any enclosed or confined fire space without proper protective equipment, including self-contained breathing apparatus (NIOSH SCBA). If possible, water should be applied as a spray from a fogging nozzle since this is a surface burning material. The application of high velocity water will spread the burning surface layer.

燃烧产物： 正常燃烧会生成二氧化碳和水蒸汽；根据燃烧温度以及空气供给量，可能会生成一氧化碳、原始单体、其它烃类物质以及烃的氧化产物。

COMBUSTION PRODUCTS:Normal combustion forms carbon dioxide, water vapor and may produce carbon monoxide, original monomer, other hydrocarbons and hydrocarbon oxidation products, depending on temperature and air availability.

第六部分 泄漏应急处置

6. ACCIDENTAL RELEASE MEASURES

意外泄漏措施：打扫干净泄漏出的物料并放入废弃物容器中。如果泄漏出的是熔体状物料，在按规定的方法进行处理前，先使其冷却、凝固。收集泄漏物，控制粉尘。不得将其倒入水源和下水道。散落的塑料粒料会引起滑跤的危险性。用清扫或真空吸除的方法清除散落的塑料粒料，将其放入桶内进行回收或废弃处置。

ACCIDENTAL RELEASE MEASURES:Sweep up material and place in a disposable container. If liquid material is spilled, allow it to cool and solidify before proceeding with disposal methods. Contain spill and control dusts. Keep out of water sources and sewers. Spilled pellets can create slipping hazard. Sweep or vacuum up spill and place in drums for recovery or disposal.

第七部分 操作处置与储存

7. HANDLING AND STORAGE

应避免加热的物料与眼睛皮肤以及衣服接触，避免吸入加热物料所散发的蒸气或烟雾。

Avoid contact of heated material with eyes, skin, and clothing. Avoidbreathing vapor or fumes from heated material.

大量聚乙烯颗粒料的散装贮存会导致有潜在爆炸可能性的乙烯（烃类）气体的积聚。应将贮存空间的乙烯（烃类）气体的浓度控制在爆炸下限浓度（LEL）2.7%以下。

Bulk storage of polyethylene pellets may result in accumulation of ethylene gas with possible explosion potential. Keep concentrations of ethylene gas below the lower explosive limit (LEL) of 2.7%.

空气中粉尘浓度超过20 mg/l后会造成粉尘爆炸的危险性。避免吸入产品的粉尘（微粉）。避免眼睛接触产品粉尘。避免皮肤和眼睛接触物料在热加工时可能散发出的废气。不得让熔融的物料与眼睛、皮肤接触或粘附在衣服上。确保加工时通风良好。操作后彻底清洗。在重新穿着前应洗涤受

到沾染的衣物。

Airborne dust concentrations above 20 mg/l may create a dust explosion hazard. Avoid inhalation of product dust (fines). Avoid eye contact with product dust. Avoid inhalation of, and eye contact with, off gases which may be produced during thermal processing. Do not get molten material in eyes, on skin or on clothing. Process only with adequate ventilation. Wash thoroughly after handling. Launder exposed clothing before reuse.

在加工设备的周围会形成物料热分解过程散发废气的凝液。在从加工设备上清除凝液时，应使用密封性手套和围裙。贮存时远离热源及火焰。

Thermal decomposition processing off gas condensate may form on surrounding equipment. Impervious gloves and apron should be used when cleaning condensate from equipment.Store away from heat or flames.

熔融的聚合物会引起严重的热灼伤。由于聚合物的导热系数较低，熔体的内部会较长时间保持灼热状态。在处置或操作这些熔体时应十分小心。散落的塑料粒料会引起滑跤的危险性。

Molten polymer may cause severe thermal burns. The interior of molten masses may remain hot for some time because of low thermal conductivity of the polymer.Use care when disposing of or handling such masses. Spilled pellets may create a slipping hazard.

避免物料污染土壤，或将这种物料排入下水道、排水系统以及水体中。

Avoid contaminating soil or releasing this material into sewage and drainage systems and bodies of water.

为了避免因静电积累而引发的危险性，在进行运输操作前应当将所有的容器和设备同导线连接好并接地。

Bond and ground all containers and equipment before transfer operations to prevent electrostatic accumulation hazard.

第八部分　接触控制/个体防护

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

一般事项：　在进行工程控制设计和选择个人防护设备时，应考虑这种物质的潜在危险性（参见第3节），这包括适用的爆炸极限、作业内容、工作地点范围内的其它物质。如果工程控制或工作惯例达不到能避免接触这种物料有害浓度水平的要求时，推荐使用以下所列的个人防护设备。由于个人防护设备只能在有限的时间和特定的情况下提供有效的防护作用，使用者应当认真阅读并充分理解随这些设备所提供的所有说明书以及限制条件。

GENERAL CONSIDERATIONS:　Consider the potential hazards of this material (see Section 3), applicable exposure limits, job activities, and other substances in the work place when designing engineering controls and selecting personal protective equipment. If engineering controls or work practices are not adequate to prevent exposure to harmful levels of this material, the personal protective equipment listed below is recommended. The user should read and understand all instructions and limitations supplied with the equipment since protection is usually provided for a limited time or under certain circumstances.

工程控制：　在通风良好的工作区域使用物料。如果加热物料时会产生蒸气或烟雾，为了控制接触，应采用密封的加工防护罩、局部通风排气以及其它工程控制措施。通风的要求应当根据现场的条件决定。如果操作会产生粉尘，为了确保粉尘的接触浓度不超过美国职业安全与卫生管理局规定的无害粉尘允许接触极限浓度（OSHA PEL），需采取特殊的通风措施。

ENGINEERING CONTROLS: Use in a well-ventilated area. If heated material generates vapor, or fumes, use process enclosures, local exhaust ventilation, or other engineering controls to control exposure. Ventilation requirements must be locally determined. If handling results in dust generation, special ventilation may be needed to ensure that dust exposure does not exceed the OSHA PEL for nuisance dust.

个人防护装备：

PERSONAL PROTECTIVE EQUIPMENT:

眼睛 / 面部防护：在操作本物料时佩戴有侧面防护罩的安全眼镜是一种良好的安全措施。如果加热这种物料，应

佩带化学防护眼镜或佩带安全眼镜和面罩。

EYE/FACE PROTECTION: Wear safety glasses with side shields when working with this material as a good safety practice. If this material is heated, wear chemical goggles or safety glasses and a face shield.

通风排气： 为了控制在推荐的接触极限浓度以下，应采用充分的通风措施。为了控制物料在热加工时会产生的废气，可能需要增加通风措施。

VENTILATION: Use adequate ventilation to control below recommended exposure levels. Supplemental ventilation may be needed to control off gases which may be produced during thermal processing.

皮肤防护： 穿戴防护服是一项良好的工业卫生作法，这可以尽可能减少与皮肤的接触。应当根据所进行的操作选择防护服。在需要操作灼热的物料时应戴上耐热手套。在从加工设备上清除物料热分解过程散发废气所形成的凝液时，应使用密封性的手套和围裙。在选择防护服时，应考虑实际要求和其它物质。

SKIN PROTECTION: Wear protective clothing to minimize skin contact as a good industrial hygiene practice. Selection of protective clothing will depend on operations conducted. Use heat resistant gloves when handling hot material. Use impervious gloves and apron when cleaning thermal decomposition processing off gas condensate from equipment.Consider physical requirements and other substanceswhen selecting protective clothing.

呼吸防护：通常不需要特殊的呼吸防护措施。在切割或磨削这些物料时，使用经英国职业安全和健康协会（IOSH）认证的口罩，该口罩适用于防护粉尘、液雾及烟雾，按时间加权平均值（TWA）测量接触极限浓度不超过 0.05 mg/m3 的情况。在无法充分地控制加热物料表面散发出的蒸汽和烟雾的情况下，应佩带经美国国立职业安全和健康研究所 / 美国矿山安全和健康监察局（NIOSH / MSHA）认证的防毒面具。

RESPIRATORY PROTECTION: No special respiratory protection is normally required. When cutting or abrading these materials use a respirator approved by IOSH for protection against dust, mists and fumes having an exposure limit measured as a time weighted average (TWA) of not less than 0.05 mg/m3.When vapors or fumes from the heated material are not adequately controlled, wear a NIOSH/MSHA approved respirator.

在使用空气净化防毒面具时选用以下滤毒元件：有机蒸汽和甲醛过滤筒。

Use the following elements for air-purifying respirators: Organic Vapor and Formaldehyde.

备注： （个人防护资料是根据通用信息而提出的，仅适用于正常的使用和条件下。如果存在特殊的、不常见的使用或条件，建议考虑寻求工业卫生学家或其它有资质的专业人士的支持。）

NOTE: Personal protection information is based upon general information as to normal uses and conditions. Where special orunusual uses or conditions exist, it is suggested that the expertassistance of an industrial hygienist or other qualified professional be sought.

第九部分 理化特性

9. PHYSICAL AND CHEMICAL PROPERTIES

物性描述： 透明、半透明的无色丸状或薄片状物质。

PH： 不适用

蒸气压： 不适用

蒸气密度（空气=1）：不适用

沸点： 不适用

熔点： 100 - 135℃

溶解度： 不溶于水

比重： 0.91 - 1.02

密度： 0.910 - 0.970 g/cm^3

蒸发速率： 无

挥发份（体积）： 0.01%

PHYSICAL DESCRIPTION:Opaque, translucent, colorless pellets or fluff

pH:	*NA*
VAPOR PRESSURE:	*NA*
VAPOR DENSITY(AIR=1):	*NA*
BOILING POINT:	*NA*
MELTING POINT:	*100 - 135 ℃*
SOLUBILITY:	*Insoluble in water*
SPECIFIC GRAVITY:	*0.91 - 1.02*
DENSITY:	*0.910 - 0.970 g/cm^3*
EVAPORATION RATE:	*NA*
PERCENT VOLATILE(VOL):	*0.01%*

第十部分　稳定性和反应性

10. STABILITY AND REACTIVITY

有害的分解产物：在热加工过程中可能生成低分子量烃类、醇类、醛类、酸类、碳氧化物及酮类物质。

HAZARDOUS DECOMPOSITION PRODUCTS:Low molecular weight hydrocarbons, alcohols, aldehydes,acids, carbon oxides and ketones can be formed during thermal processing.

化学稳定性：　稳定。

CHEMICAL STABILITY: Stable.

应避免的条件：避免加热到推荐的加工温度以上。不得加热到自燃温度以上。禁止在无良好通风的情况下进行加热。

CONDITIONS TO AVOID: Avoid heating above the recommended processing temperature. Do not heat above the autoignition temperature. Do not heat without adequate ventilation.

与其它物质的不相容性：可与氯酸盐、硝酸盐及过氧化物等强氧化剂发生反应。可与游离态卤素发生反应。

INCOMPATIBILITY WITH OTHER MATERIALS:May react with strong oxidizing agents, such as chlorates, nitrates, peroxides, etc. May react with free halogens.

危险的聚合反应：聚合反应不会发生。

HAZARDOUS POLYMERIZATION: Polymerization will not occur.

第十一部分　毒理学资料

11. TOXICOLOGICAL INFORMATION

对眼睛的影响：对眼睛刺激性危害是根据类似物质的数据提出的。

EYE EFFECTS:The eye irritation hazard is based on data for a similar material.

对皮肤的影响：对皮肤刺激性危害是根据类似物质的数据提出；急性皮肤毒性是根据类似物质的数据提出的。

SKIN EFFECTS: The skin irritation hazard is based on data for a similar material. The acute dermal toxicity is based on data for a similar material.

急性口服毒性：急性口服毒性是根据类似物质的数据提出的。

ACUTE ORAL EFFECTS: The acute oral toxicity is based on data for a similar material.

急性吸入效应：急性呼吸毒性是根据类似物质的数据提出的。

ACUTE INHALATION EFFECTS: The acute respiratory toxicity is based on data for a similar material.

其它毒物学信息：在一项以人为受检者的皮肤过敏性试验研究中，类似物质没有引起皮肤过敏。

ADDITIONAL TOXICOLOGY INFORMATION: A similar material did not cause skin sensitization in a patch test study with human subjects.

聚乙烯在热加工过程中能够降解。控制其降解的三个变量为温度、在该温度下的持续时间和供氧量。根据具体加工条件，会形成各种各样的低分子量烃类、醇类、醛类、酸类及酮类物质。这些降解产物均是呼吸的刺激物。已经显示长期和反复地吸入烟雾成分会造成其它不利于健康的影响。应当遵循本材料安全数据表（MSDS）中提出的所有建议，尽可能减少与加工排放物接触。

This product contains polymerized ethylene. During thermal processing, this polymer can degrade. The three variables which control its degradation are the temperature, the length of time at that temperature, and the amount of oxygen available. Depending on the local processing conditions, a variety of low molecular weight hydrocarbons, alcohols, aldehydes, acids, and ketones can be formed. These materials are respiratory irritants. Prolonged and repeated breathing of fume components has been shown to cause other adverse health effects. Exposure to processing emissions should be minimized by following all recommendations in this MSDS.

第十二部分　生态学资料
12. ECOLOGICAL INFORMATION

生态毒性：　鱼类或鸟类有可能吞咽塑料颗粒，这样会堵塞它们的消化道。

ECOTOXICITY: Fish or birds may eat pellets which may obstruct their digestive tracts.

环境后果：　本物质不容易被生物降解。

ENVIRONMENTAL FATE: This material is not expected to be readily biodegradable.

第十三部分　废弃处置
13. DISPOSAL CONSIDERATIONS

将受到污染的物料置入一次性的容器内，按照符合可适用法规的方式进行废弃处置。

Place contaminated materials in disposable containers and dispose of in a manner consistent with applicable regulations.

与当地环境或健康管理机构联系，得到废弃处置这种物料的批准。回收物料，以便重新使用、循环利用、焚烧回收能量或放入废弃物管理设施。在进行废弃处置前，向您的环境保护联系人咨询，决定是否要进行 TCLP 试验（毒性特征淋浙试验方法，美国环保总署（EPA）试验方法 1311）。参考 40 CFR 法规的第 261 节。

Contact local environmental or health authorities for approved disposal of this material. Recover for reuse,.recycle, incinerate for energy or place in waste management facility. Prior to disposal, consult your environmental contact to determine if the TCLP (Toxicity Characteristic Leaching Procedure, EPA Test Method 1311) is required. Reference 40 CFR Part 261.

第十四部分　运输信息
14. TRANSPORT INFORMATION

所给出的说明可能不适用于所有的运输状况。如有另外的说明要求（例如技术名称）和特定运输模式及特定数量的运输要求，请查询法规 49 CFR，或适用的货物运输法规。

The description shown may not apply to all shipping situations. Consult 49CFR, or appropriate Dangerous Goods Regulations, for additional description requirements (e.g., technical name) and mode-specific or quantity-specific shipping requirements.

美国运输部（DOT）运输货名：　美国运输部（DOT）不将该产品指定为危险性物质。

DOT SHIPPING NAME: NOT DESIGNATED AS A HAZARDOUS MATERIAL BY THE FEDERAL DOT.

美国运输部（DOT）危险品分类：　不适用

DOT HAZARD CLASS: NOT APPLICABLE

美国运输部（DOT）识别编号：　不适用

DOT IDENTIFICATION NUMBER: NOT APPLICABLE

美国运输部（DOT）包装类别：　不适用

DOT PACKING GROUP: NOT APPLICABLE

第十五部分　法规信息
15. REGULATORY INFORMATION

濒危物种法案（SARA） 311 分类：

1. 立即的(急性)健康影响： 无
2. 延缓的 (慢性) 健康影响： 无
3. 火灾危险性： 无
4. 突然性的危险性： 无
5. 反应性危险： 无

SARA 311 CATEGORIES:

1. Immediate (Acute) Health Effects: NO

2. Delayed (Chronic) Health Effects: NO

3. Fire Hazard: NO

4. Sudden Release of Pressure Hazard: NO

5. Reactivity Hazard: NO

第十六部分 其他信息

16. OTHER INFORMATION

全美消防协会（NFPA）分级： 健康 1； 易燃性 1； 反应性 0。

危险性物质识别体系（HMIS）分级： 健康 0； 易燃性 1； 反应性 0。

（0-最小、 1-轻微、2-中等、3-高度、 4-极其；PPE：个人防护设备指标推荐；*- 慢性效应标志）。

NFPA RATINGS: Health 1; Flammability 1; Reactivity 0;

HMIS RATINGS: Health 0; Flammability 1; Reactivity 0;

*(0-Least, 1-Slight, 2-Moderate, 3-High, 4-Extreme, PPE:- Personal Protection Equipment Index recommendation, *- Chronic Effect Indicator).*

这些数值是使用由全美消防协会(NFPA)或全美油漆和涂料协会所编制的指南及发表的评价报告而获取(用于危险性物质识别体系 HMIS 分级)。

These values are obtained using the guidelines or published evaluations prepared by the National Fire Protection Association (NFPA) or the National Paint and Coating Association.(for HMIS ratings).

“HHMTR210”系列产品的添加剂中含有“抗铜剂”，“HHMTR144”系列的“HHMTR144P”牌号的产品可能含有“抗铜剂”。因而，关于上述产品的物料安全数据，除了本物料安全数据表已列出的内容外，还应参考有关“抗铜剂”的物料安全数据。

HHMTR210 series products contain copper-inhibitor. HHMTR144P among HHMTR144 series products may contain copper-inhibitor. Therefore, except for MSDS provided herein, MSDS for copper-inhibitor shall be referred to too.

本“MSDS”依据本公司生产专利技术提供方（美国雪佛龙菲利普斯化学公司）提供的“MSDS”编制。

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